COMPARISON OF SELF LEARNING AGENTS

using RL and IL in unreal engine 5

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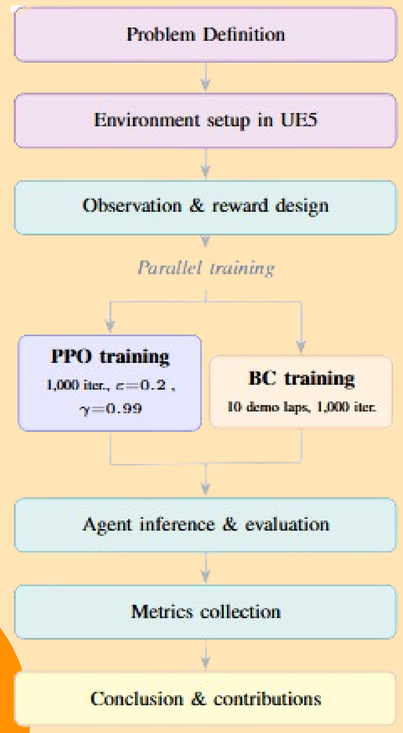
Research Overview

This research examines how different machine learning paradigms Reinforcement Learning and Imitation Learning affect autonomous agent performance in a simulated driving environment built on Unreal Engine 5.

Objective

To compare PPO (RL) and Behavior Cloning (IL) agents on the same closed-loop racing track, analyzing lap completion rate, lap time, and convergence behavior to identify the strengths and trade-offs of each paradigm.

Methodology Flowchart



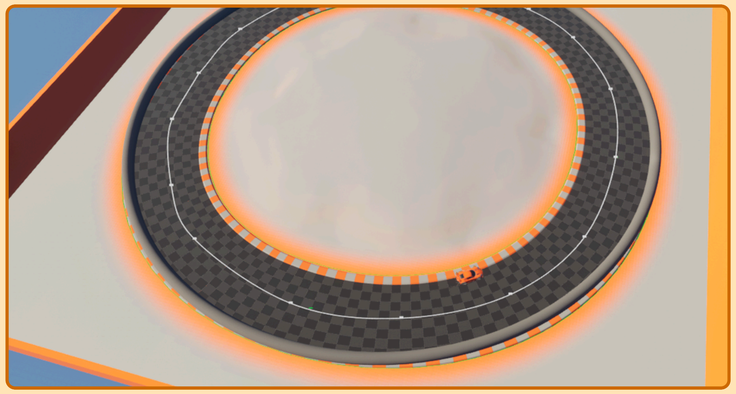
Methodology Overview

Environment Setup
The simulation is built in Unreal Engine 5.6.1 using the Learning Agents plugin, featuring a closed-loop racing track built via a Spline component and realistic vehicle dynamics powered by UE5's Chaos Vehicle Physics system.

Agent Training
PPO is trained using 8 parallel agents over 1,000 iterations (ε = 0.2, γ = 0.99), rewarding velocity alignment along the spline and penalizing lateral deviation beyond 800 cm. BC is trained from 10 expert demonstration laps recorded by a human operator, requiring no reward function.

Observation Space
Both agents observe vehicle position along the spline, heading relative to the spline tangent, and current velocity. Since PPO uses 8 parallel agents, BC is given one additional observation, a vehicle yaw angle observation to balance the training asymmetry.

Evaluation
Both agents are evaluated over 200 inference episodes under identical conditions. An episode is complete when the vehicle crosses the finish line, and failed if stationary for more than 30 seconds. Performance is measured via lap completion rate, mean lap time, and cumulative reward (PPO only).

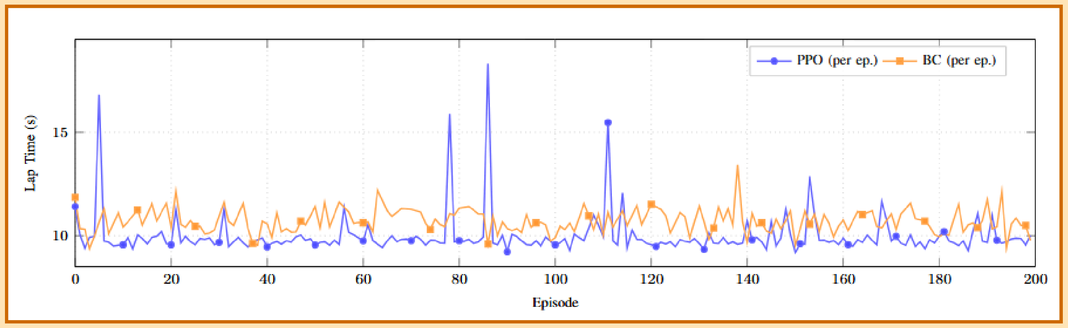


(Enviroment of Agent Evaluation)

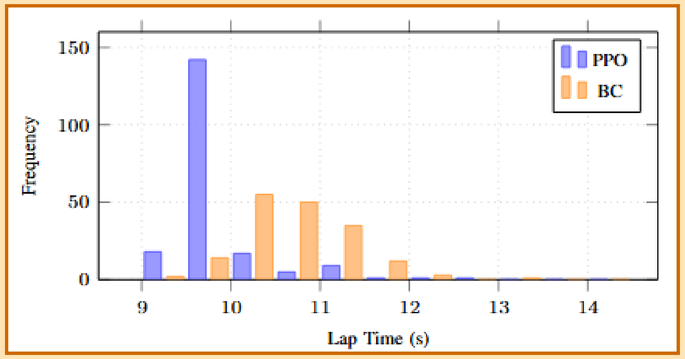
Result Table

Metric	PPO (RL)	BC (IL)
Lap Completion Rate (%)	99	86
Failed Episodes	2	28
Mean Cumulative Reward (s)	0.464 ± 0.045	-
Mean Lap Time (s)	9.99 ± 1.08	10.68 ± 0.59

Data Highlight



(Chart A)



(Chart B)

Chart Information

Chart A
Comparison of lap times per episode between PPO and BC, showing PPO's more consistent performance and lower failure rate.

Chart B
Lap time frequency distribution for PPO and BC, highlighting PPO's strong concentration around the optimal lap time range.

Conlusion & Future Works

Conclusion
PPO outperforms BC in both lap completion rate (99% vs 86%) and mean lap time (9.99s vs 10.68s), with the difference statistically significant ($t = -7.66$, $p \ll 0.001$). BC demonstrates greater lap-to-lap consistency (σ = 0.59s) but is fundamentally limited by its demonstration data, when the vehicle encounters unseen states, covariate shift causes unrecoverable failures. UE5 Learning Agents plugin is confirmed as a viable platform for comparative ML research

Future Works
Future work includes replacing pure BC with DAgger to mitigate covariate shift, evaluating both agents on unseen track configurations, and exploring hybrid RL–IL approaches where BC initializes the PPO policy before fine-tuning. Incorporating richer sensor inputs such as LiDAR-like raycasts is also identified as a direction toward more realistic autonomous driving conditions.